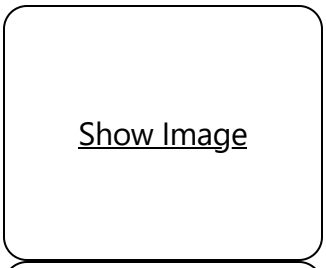
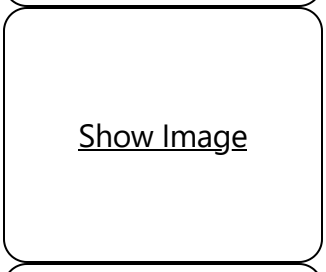
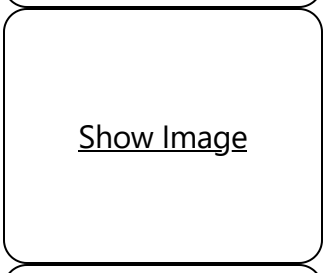
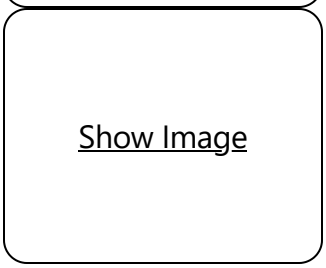
AI-Powered Personal Knowledge Assistant









> A privacy-first, offline-capable AI assistant that integrates with your personal data to provide hyper-personalized insights, task prioritization, and decision-making support. Unlike cloud-based AIs, it runs entirely on your local device for maximum privacy and control.

🌟 Features

🔒 Privacy-First Architecture

- **100% Local Processing**: All AI computations happen on your device
- **Zero Cloud Dependency**: No data ever leaves your machine
- **Advanced Encryption**: Local data storage with AES-256 encryption
- **Open Source**: Full transparency in data handling and AI algorithms
- **GDPR Compliant**: Complete data sovereignty and right to deletion

🧠 Intelligent Data Integration

- **Multi-Source Connectivity**: Notes, emails, calendar, documents, contacts, photos

- **Smart Indexing**: Vector embeddings for semantic search across all data types
- **Real-Time Sync**: Automatic updates when your data changes
- **Format Agnostic**: Supports text, images, PDFs, office documents, and more
- **Relationship Mapping**: Discovers connections between different data sources

💡 AI-Powered Insights

- **Pattern Recognition**: Identifies productivity patterns and behavioral insights
- **Knowledge Clustering**: Groups related information across different sources
- **Trend Analysis**: Tracks changes in your work patterns over time
- **Predictive Suggestions**: Anticipates your needs based on historical data
- **Anomaly Detection**: Flags unusual patterns that might need attention

✅ Smart Task Management

- **Context-Aware Prioritization**: Uses email urgency, calendar conflicts, and deadlines
- **Dynamic Re-ranking**: Adjusts priorities as new information arrives
- **Deadline Intelligence**: Predicts realistic completion times based on your history
- **Dependency Mapping**: Identifies task relationships and blockers
- **Workload Balancing**: Distributes tasks across optimal time slots

💬 Natural Language Interface

- **Conversational AI**: Chat naturally about your data and tasks
- **Multi-Modal Queries**: Ask questions using text, voice, or images
- **Context Retention**: Remembers conversation history for better responses
- **Command Recognition**: Understands complex, multi-step requests
- **Explanation Mode**: Provides reasoning behind suggestions and insights

🚀 Quick Start

Prerequisites

```bash
# System Requirements
- RAM: 8GB minimum, 16GB recommended
- Storage: 5GB free space for models and index
- OS: Windows 10+, macOS 10.15+, Ubuntu 18.04+
- GPU: Optional but recommended for faster processing
```

Installation

```bash
# Clone the repository
git clone https://github.com/yourusername/personal-ai-assistant.git
cd personal-ai-assistant

# Install dependencies
npm install

# Download AI models (first run only)
npm run setup-models

# Start the application
npm start
```

First-Time Setup

1. **Data Source Configuration**

   ```bash
   # Configure your data sources
   npm run config
   ```

2. **Initial Indexing**

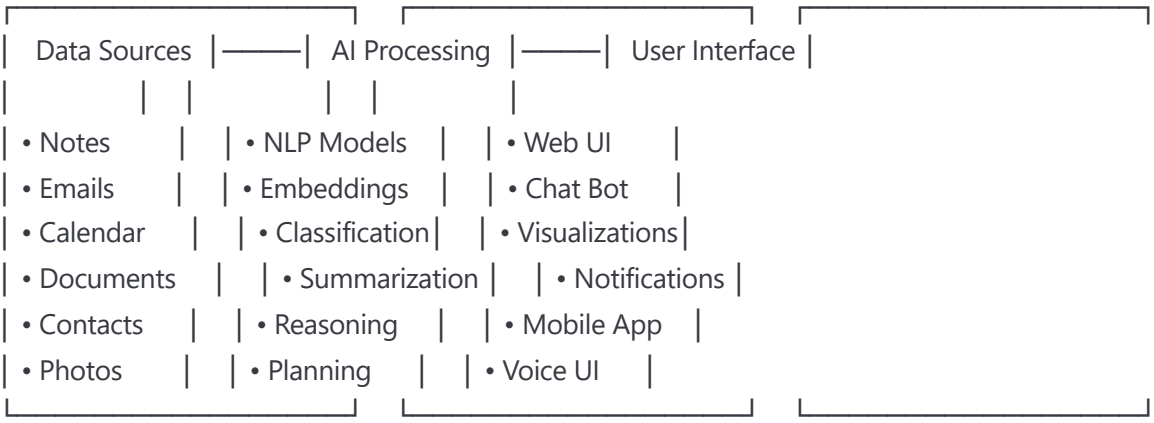
   ```bash
   # Index your existing data (may take 10-30 minutes)
   npm run index-data
   ```

3. **Security Setup**

```bash
# Set up encryption keys
npm run setup-security
```

📊 Architecture Overview

Core Components



AI Models Stack

Component	Model	Purpose	Size
Language Understanding	Llama 2 7B (Quantized)	Text processing, chat	4GB
Embeddings	E5-Large	Semantic search, clustering	1.2GB
Classification	DistilBERT	Email/task categorization	250MB
Summarization	BART-Large	Document summarization	1.6GB
Vision	CLIP	Image understanding	350MB

Data Flow

```mermaid
graph TD
    A[Personal Data Sources] --> B[Data Ingestion Layer]
    B --> C[Preprocessing & Cleaning]
    C --> D[Vector Embedding Generation]
    D --> E[Local Vector Database]
    E --> F[AI Analysis Engine]
    F --> G[Insight Generation]
    G --> H[User Interface]
    H --> I[User Interaction]
    I --> F
```

🔧 Configuration

Data Sources Setup

Create a `config/data-sources.json` file:

```json
{
  "notes": {
    "enabled": true,
    "paths": [
      "~/Documents/Notes",
      "~/Obsidian Vault"
    ],
    "formats": ["md", "txt", "org"],
    "scan_interval": "5m"
  },
  "emails": {
    "enabled": true,
    "type": "imap",
    "settings": {
      "host": "localhost",
      "port": 993,
      "secure": true,
      "auth": {
        "user": "encrypted_user",
        "pass": "encrypted_pass"
      }
    }
  },
  "calendar": {
    "enabled": true,
    "type": "caldav",
    "url": "https://your-calendar-server.com/cal.php/calendars/user/"
  }
}
```

AI Model Configuration

Edit `config/ai-models.json`:

json


```
{
  "language_model": {
    "name": "llama2-7b-chat",
    "quantization": "q4_0",
    "context_length": 4096,
    "temperature": 0.7
  },
  "embedding_model": {
    "name": "e5-large-v2",
    "dimensions": 1024,
    "batch_size": 32
  },
  "processing": {
    "max_concurrent_tasks": 4,
    "gpu_acceleration": true,
    "memory_limit": "6GB"
  }
}
```


🎛️ Usage Examples

Basic Queries

```
# Find information across all sources
"What did I write about machine learning last month?"

# Task prioritization
"What should I work on today?"

# Meeting preparation
"Summarize everything related to the Johnson project for tomorrow's meeting"

# Pattern analysis
"When am I most productive during the week?"
```

Advanced Workflows

```
# Complex analysis
"Analyze my email patterns with the marketing team and suggest optimal meeting times"

# Predictive insights
"Based on my calendar, when will I have time for the quarterly report?"

# Cross-reference queries
"Find all notes, emails, and documents related to budget planning"

# Productivity optimization
"What tasks am I consistently procrastinating on and why?"
```

🛠️ Development

Project Structure

```
personal-ai-assistant/
├── src/
│   ├── core/              # Core AI processing
│   │   ├── models/        # AI model interfaces
│   │   ├── processing/    # Data processing pipelines
│   │   └── reasoning/     # Decision-making logic
│   ├── data/              # Data ingestion and management
│   │   ├── connectors/    # Data source connectors
│   │   ├── indexing/      # Vector database operations
│   │   └── storage/       # Local storage management
│   ├── ui/                # User interface components
│   │   ├── web/           # Web interface
│   │   ├── mobile/        # Mobile app
│   │   └── cli/           # Command-line interface
│   └── security/          # Security and encryption
├── models/                # Downloaded AI models
├── config/                # Configuration files
├── data/                  # User data and indexes
└── docs/                  # Documentation
```

Building from Source

```bash
# Install development dependencies
npm install --dev

# Run tests
npm test

# Build for production
npm run build

# Create distributable packages
npm run package
```

Contributing

1. Fork the repository
2. Create a feature branch: `git checkout -b feature/amazing-feature`
3. Make your changes and add tests
4. Commit: `git commit -m 'Add amazing feature'`
5. Push: `git push origin feature/amazing-feature`
6. Open a Pull Request

🔐 Security & Privacy

Data Protection

- **Local Storage Only**: All data remains on your device
- **Encryption at Rest**: AES-256 encryption for all stored data
- **Secure Communication**: TLS 1.3 for any local network communication
- **Access Controls**: Biometric/password protection for sensitive data
- **Audit Logging**: Complete log of data access and modifications

Privacy Guarantees

- **No Telemetry**: Zero usage data collection
- **No Cloud Processing**: All AI inference happens locally
- **Anonymous Usage**: No user identification or tracking
- **Data Sovereignty**: Complete control over your information

- **Right to Deletion**: Easy data removal and account deletion

Security Best Practices

```bash
# Enable security features
npm run enable-security

# Regular security updates
npm run security-update

# Backup encrypted data
npm run backup --encrypted

# Verify data integrity
npm run verify-integrity
```

📈 Performance & Optimization

System Requirements

Configuration	Minimum	Recommended	Optimal
RAM	8GB	16GB	32GB
CPU	4 cores	8 cores	16 cores
Storage	10GB	50GB	100GB SSD
GPU	None	GTX 1060	RTX 3080+

Performance Tuning

```bash
# Optimize for your hardware
npm run optimize-performance

# Monitor resource usage
npm run monitor

# Profile AI model performance
npm run profile-models
```

🧪 Advanced Features

Plugin System

Create custom data connectors:

```javascript
// plugins/custom-source.js
class CustomDataSource {
  async connect(config) {
    // Implementation
  }

  async fetchData() {
    // Data retrieval logic
  }

  async processData(data) {
    // Data processing
  }
}

module.exports = CustomDataSource;
```

API Integration

```javascript
// Enable REST API for external integrations
const assistant = require('./src/core/assistant');

assistant.startAPI({
  port: 3000,
  auth: 'bearer-token',
  cors: false  // Local only
});
```

Automation Scripts

```bash
# Schedule automatic insights
crontab -e
0 9 * * * /usr/local/bin/assistant generate-daily-insights

# Backup routine
0 2 * * 0 /usr/local/bin/assistant backup --encrypt
```

🚀 Deployment Options

Desktop Application

```bash
# Build native desktop app
npm run build:desktop
```

Mobile App

```bash
# Build mobile applications
npm run build:mobile:ios
npm run build:mobile:android
```

Server Deployment (Family/Team)

```bash
# Deploy on local network server
npm run deploy:server
```

📚 Additional Resources

- API Documentation
- Plugin Development Guide
- Troubleshooting
- Performance Tuning
- Security Guide
- Community Forum

🤝 Support

- **Documentation**: docs.personal-ai.dev
- **Community**: Discord Server
- **Issues**: GitHub Issues
- **Email**: support@personal-ai.dev (for security issues only)

📄 License

This project is licensed under the MIT License - see the LICENSE file for details.

🏆 Acknowledgments

- **Hugging Face** for open-source AI models
- **Onnx Runtime** for efficient model inference
- **Electron** for cross-platform desktop support
- **Community Contributors** for feedback and improvements

Built with ❤️ for privacy-conscious users who want intelligent assistance without compromising their data

⭐ Star this repo • 🐛 Report Bug • 💡 Request Feature